

Technology #139

Hybrid thermal-mechanical treatment using
localized heating during rail joint welding
while simultaneously applying crack-resistant
material inserts

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Hybrid thermal-mechanical treatment using localized heating during rail joint welding while simultaneously applying crack-resistant material inserts

WHY DOES IT MATTER?

Documented recurring problem 'rail steel fatigue fracture' (broken rails/joints, FRA cause family T2): 3 of 25 FRA Form 54 accidents in the retrieved window are broken-rail/joint-caused, derailling 17 cars total. Causes observed: 'Broken Rail - Base [Desc.Changed-Refer to Current/Prev.Guides]'; 'Mismatched rail-head contour'. LIMITS STAMPED INLINE: carrier-reported regulatory reports; cause codes are classifications of the initial report, NOT proven root causes; reportable-above-threshold events only (Art. XXI.5 analog).

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in thermal. 9-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 9-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Comparative testing of conventional thermite welds versus hybrid thermal-mechanical treated joints under hydrogen exposure and cyclic loa.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: temperature-rise mapping at max duty. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden; interface contact resistances dominate at small scale and must be measured, not assumed zero; steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k; one-dimensional gradient (lateral gradients neglected); TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.; TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.; TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.; Model input variables requiring sourced values before any numeric result: h ($W/(m^2.K)$), A (m^2), dT (K), L_i (m), k_i ($W/(m.K)$), A_i (m^2), k ($W/(m.K)$), A (m^2), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , $W/(m.K)$] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m^2] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Thermite welding combined with hydrogen-resistant material inserts to prevent embrittlement and enhance fracture toughness

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'rail steel fatigue fracture' (broken rails/joints, FRA cause family T2): 3 of 25 FRA Form 54 accidents in the retrieved window are broken-rail/joint-caused, derailling 17 cars total. Causes observed: 'Broken Rail - Base [Desc.Changed-Refer to Current/Prev.Guides]'; 'Mismatched rail-head contour'. LIMITS STAMPED INLINE: carrier-reported regulatory reports; cause codes are classifications of the initial report, NOT proven root causes; reportable-above-threshold events only (Art. XXI.5 analog).

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: thermal

Disciplines: heat transfer, thermal engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Hybrid thermal-mechanical treatment using localized heating during rail joint welding while simultaneously applying crack-resistant material inserts

Subsystems:

?: heat source characterization - ?

?: conduction path / spreader - ?

?: reject interface - ?

?: safety limit interlock - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Hybrid thermal-mechanical treatment using localized heating during rail joint welding while simultaneously applying crack-resistant material inserts

Key Physics: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: dT .

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected). Engaged variables: k , dT/dx .

Assumptions:

- h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden
- interface contact resistances dominate at small scale and must be measured, not assumed zero
- steady-state conduction (no transient term)
- isotropic, homogeneous medium conductivity k
- one-dimensional gradient (lateral gradients neglected)
- TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.
- TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.
- TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.

- Model input variables requiring sourced values before any numeric result: h (W/(m².K)), A (m²), dT (K), L_i (m), k_i (W/(m.K)), A_i (m²), k (W/(m.K)), A (m²), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , W/(m.K)] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m²] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'rail steel fatigue fracture in service', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'hotspot exceeding material/safety limit', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'interface delamination raising resistance over life', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coolant/flow path blockage (if active)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'thermal runaway coupling to electronics', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'condensation at cold surfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Detection or material design must prevent service fracture under measured thermal stresses across temperature excursions;	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Thermite welding combined with hydrogen-resistant material inserts to prevent embrittlement and enhance fracture toughness.	MODIFIED
DI-003	Expected effect target	Prevention of hydrogen absorption during welding and operation, maintaining structural integrity under temperature excursions.	MODIFIED
DI-004	Domain design input pattern: dissipation applicability NOT ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING_PROPOSED	
DI-005	Domain design input pattern: ambient application NOT ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING_PROPOSED	
DI-006	Domain design input pattern: safety application [ED best practices] "temperature"	ENGINEERING_PROPOSED	
DI-007	Regulatory design input: compliance candidate standards (e.g., ASME BPE, IEC 60076-4) (general electrical safety UNKOWN) (PRELIMINARY PROPOSAL)	is never inferred	
DI-008	Custodied observation: Mechanical properties and density of materials (e.g., 7050-T6 aluminum) (TC4 AcBoville5308J8302592atf06db212e4b465a9fdb3e7d)		
DI-009	Custodied observation: Advances in Resistant Welding of Fiberglass Composites (66ef3f10ab6c36dd8f0e05e581dff8a548ba3ad8a14a09f)		
DI-010	Custodied observation: Non-Destructive Testing and Evaluation of Hydrogen Absorption in Advanced Structural Components (70e24413baf0e0260f0e0265285eb2a1a06a4366bf9a062)		
DI-011	Custodied observation: Current research/design progress and contributions (9a0df62945e58fa02074bb4b04b5fb000e7670f1fba)		
DI-012	Custodied observation: Evaluation of Reliability Steel Thermite Wires (415157d19geonDegraded0b19pf54278d019f00008ac07ca66c917b4d32af2b1f195)		

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	heat source characterization realizing Hybrid thermal-mechanical treatment using localized heating during rail joint welding while simultaneously applying	CONCEPTUAL	localized heating definition
DO-002	conduction path / spreader realizing Hybrid thermal-mechanical treatment using localized heating during rail joint welding while simultaneously applying	CONCEPTUAL	localized heating definition
DO-003	reject interface realizing Hybrid thermal-mechanical treatment using localized heating during rail joint welding while simultaneously applying crack-resistant	CONCEPTUAL	localized heating definition
DO-004	safety limit interlock realizing Hybrid thermal-mechanical treatment using localized heating during rail joint welding while simultaneously applying crack-resistant	CONCEPTUAL	localized heating definition
DO-005	heat source characterization interfaces with conduction path / spreader: signal/energy/material flow off (for hybrids, not a mechanical treatment using localized heating)	CONCEPTUAL	energy/material flow off (for hybrids, not a mechanical treatment using localized heating)
DO-006	conduction path / spreader interfaces with reject interface: signal/energy/material flow off (for hybrids, not a mechanical treatment using localized heating)	CONCEPTUAL	energy/material flow off (for hybrids, not a mechanical treatment using localized heating)
DO-007	reject interface interfaces with safety limit interlock: signal/energy/material flow off (for hybrids, not a mechanical treatment using localized heating)	CONCEPTUAL	energy/material flow off (for hybrids, not a mechanical treatment using localized heating)
DO-008	Device boundaries: patient/fluid inlet and outlet termination/CONTROL/control/data/parameters retained by system thermal definitions	CONCEPTUAL	parameters retained by system thermal definitions
DO-009	Flow/force/signal path geometry realizing the mechanical treatment using localized heating during rail joint welding while simultaneously applying	UNKNOWN	mechanical treatment using localized heating
DO-010	Operating range for heat generation at source within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-011	Operating range for spreader conductivity within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-012	Operating range for interface contact resistance within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-013	Operating range for reject-path area within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-014	Operating range for allowable temperature rise within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-015	Operating range for ambient worst case within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-016	Sensing -> decision -> actuation loop maintaining the mechanical treatment using localized heating during rail joint welding while simultaneously applying	REPOSES	operating position, hybridity, selection, decision thresholds, localize values required
DO-017	Information path: sensing element -> decision/monitor	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical information
DO-018	Fixture enabling: temperature-rise mapping at max duty	UNKNOWN	concept named, build specification, instrumentation selection; acceptance instrumentation
DO-019	Fixture enabling: interface resistance characterization	UNKNOWN	concept named, build specification, instrumentation selection; acceptance instrumentation
DO-020	Fixture enabling: worst-case ambient soak test — rig concept	UNKNOWN	concept named, build specification, instrumentation selection; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: temperature-rise mapping at maximum	UNKNOWN	Unknown concept name; fixture specific instrumentation selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
heat generation at source	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
spreader conductivity	UNKNOWN (no sourced value)	W/(m.K)	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
interface contact resistance	UNKNOWN (no sourced value)	K/W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
reject-path area	UNKNOWN (no sourced value)	m^2	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
allowable temperature rise	UNKNOWN (no sourced value)	K	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
ambient worst case	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
rail steel fatigue fracture	Fails by rail steel fatigue fracture	design service NOF test data (NOF test data requires design work)	NOF test data (NOF test data requires design work)	UNKNOWN (no sourced value) and verification method L
hotspot exceeding material safety limit	Fails by hotspot exceeding material safety limit	spread safety limit; heat-seal spreading gutters; heat-seal spreading gutters; heat-seal spreading gutters; heat-seal spreading gutters	spread safety limit; heat-seal spreading gutters; heat-seal spreading gutters; heat-seal spreading gutters; heat-seal spreading gutters	UNKNOWN (no sourced value) and verification method L
interface delamination	Fails by interface delamination	Massing resistance over life cycle; Massing resistance over life cycle; Massing resistance over life cycle; Massing resistance over life cycle	Massing resistance over life cycle; Massing resistance over life cycle; Massing resistance over life cycle; Massing resistance over life cycle	UNKNOWN (no sourced value) and verification method L
coolant/flow path blockage	Fails by coolant/flow path blockage	design policy; coolant/flow path blockage; coolant/flow path blockage; coolant/flow path blockage	design policy; coolant/flow path blockage; coolant/flow path blockage; coolant/flow path blockage	UNKNOWN (no sourced value) and verification method L
thermal runaway coupling	Fails by thermal runaway coupling	design policy; thermal runaway coupling; thermal runaway coupling; thermal runaway coupling	design policy; thermal runaway coupling; thermal runaway coupling; thermal runaway coupling	UNKNOWN (no sourced value) and verification method L
condensation at cold surfaces	Fails by condensation at cold surfaces	design policy; condensation at cold surfaces; condensation at cold surfaces; condensation at cold surfaces	design policy; condensation at cold surfaces; condensation at cold surfaces; condensation at cold surfaces	UNKNOWN (no sourced value) and verification method L

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
rail steel fatigue fracture in service	Fails by rail steel fatigue fracture in service	designer did not establish a baseline of high cycle fatigue properties for the rail steel (Abs X&Y)	Uncertainty in high cycle fatigue properties (Abs X&Y)
hotspot exceeding material safety limit	Fails by hotspot exceeding material/safety limit	pylons/gear heavy gear station on speed high cycle fatigue properties (Abs X&Y)	Uncertainty in high cycle fatigue properties (Abs X&Y)
interface delamination raising risk of structural failure	Fails by structural delamination raising risk of structural failure	EB selective real time cyclic expansion cycle life with 100% inspection of structural delamination (Abs X&Y)	Uncertainty in high cycle fatigue properties (Abs X&Y)
coolant/flow path blockage (if active)	Fails by coolant/flow path blockage (if active)	passive flow path blockage (if active) passive flow path blockage (if active) passive flow path blockage (if active)	Uncertainty in high cycle fatigue properties (Abs X&Y)
thermal runaway coupling	Fails by thermal runaway coupling	electronic safety temperature analysis (if active) electronic safety temperature analysis (if active)	Uncertainty in high cycle fatigue properties (Abs X&Y)
condensation at cold surfaces	Fails by condensation at cold surfaces	surface coating with low thermal conductivity (if active) surface coating with low thermal conductivity (if active)	Uncertainty in high cycle fatigue properties (Abs X&Y)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Comparative testing of conventional thermal analysis	Comparative testing of conventional thermal analysis	Comparative testing of conventional thermal analysis	Comparative testing of conventional thermal analysis
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	temperature-rise mapping at max duty	Acceptance criteria (ENGINERING_PROPOSAL) precision fixed	Acceptance criteria (ENGINERING_PROPOSAL) precision fixed	Acceptance criteria (ENGINERING_PROPOSAL) precision fixed
VF-004	temperature-rise mapping at maximum	temperature-rise mapping sensing (one played) (if active) temperature-rise mapping sensing (one played) (if active)	temperature-rise mapping sensing (one played) (if active) temperature-rise mapping sensing (one played) (if active)	temperature-rise mapping sensing (one played) (if active) temperature-rise mapping sensing (one played) (if active)
VF-005	thermal-resistance trend over acceleration	thermal-resistance trend over acceleration	thermal-resistance trend over acceleration	thermal-resistance trend over acceleration
VF-006	temperature-rise mapping at max duty	Acceptance criteria (ENGINERING_PROPOSAL) precision fixed	Acceptance criteria (ENGINERING_PROPOSAL) precision fixed	Acceptance criteria (ENGINERING_PROPOSAL) precision fixed
VF-007	worst-case ambient soak with staged power	worst-case ambient soak with staged power	worst-case ambient soak with staged power	worst-case ambient soak with staged power

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	Acceptance criteria to be fixed	Validation criteria to be fixed

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	biocompatible thermal spreader (graphite/AlN)	graphite/AlN	electronics thermal practice	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
biocompatible thermal spreader	biocompatible thermal spreader	1	STANDARD	biocompatible thermal spreader	Not Established	STANDARD	Stability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- bonded spreader + encapsulation: domain registry manufacturing route - candidate only; no process qualification exists
- thermal interface material application control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter heat generation at source [q W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter spreader conductivity [k_i W/(m.K)]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter interface contact resistance [R_total K/W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter reject-path area [A m^2]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter allowable temperature rise [dT K]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter ambient worst case [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}

- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification	1050 Hz frequency response	NOT ESTABLISHED	1050 Hz frequency response	1050 Hz frequency response
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance	constraint-compliance (to be quoted)
WP-03	article for VF-001	per method	Comparative testing	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment:	NOT ESTABLISHED	VF-002 verification	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	temperature-rise mapping	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	temperature-rise mapping	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	thermal-resistance trend	NOT ESTABLISHED	VF-005 verification	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	temperature-rise mapping	NOT ESTABLISHED	VF-006 verification	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	worst-case ambient	NOT ESTABLISHED	VF-007 verification	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 9-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.